

Chapter 78

Automation & Integration Question Bank

Choose the right automation approach for a given problem, macro, script, low-code tool, or RPA, instead of defaulting to whichever one you know best · design report and alert automations that actually get read · reason correctly about APIs, webhooks, and reverse ETL

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Chapter 78. Automation & Integration

Question Bank

Part VIII — The Interview Playbook

You will learn to: choose the right automation approach for a given problem, macro, script, low-code tool, or RPA, instead of defaulting to whichever one you know best · design report and alert automations that actually get read · reason correctly about APIs, webhooks, and reverse ETL · build retry and idempotency logic that survives a flaky upstream system · investigate an automation that silently stopped working, the way an interviewer actually wants.

How this chapter is built. Same format as Chapters 70–77: every core question leads with a "Remember it as..." hook, a one-line answer, a compact tier table. Rapid-fire sections are scan tables. **Every Python pattern in this chapter was actually executed**, not described from memory: retry-with-backoff, webhook idempotency, rate limiting, and upsert-based CRM sync are all shown with real, run output. Section order was planned before writing: choosing a tool first, then the specific automation types (reports, APIs, sync), then failure handling, then full design cases pulling it together.

Learn it in pointers reference Chapter 19 (VBA/Apps Script automation) and Chapter 70 (§70.5–70.6, the same automation platforms tested from a different angle) at chapter level for Chapter 19, since this chat doesn't have its approved text to check exact section numbers against.

78.1 Choosing the right automation approach

Q78-001 · A stakeholder wants a manual weekly task automated. Walk through how you'd choose between a macro, a script, a low-code tool, and RPA

Remember it as: The right tool depends on how the task touches its systems: inside one app (macro), across systems with APIs (script), across systems with a friendly builder and no APIs needed (low-code), or across systems with no APIs at all, mimicking clicks (RPA), each one a step further from "clean" and a step closer to "works on anything."

Answer in one line: A **macro** (VBA, Apps Script) suits automation entirely inside one application; a **script** (Python, calling APIs directly) suits automation across systems that expose clean APIs; a **low-code tool** (Zapier, Power Automate) suits cross-system automation where a visual builder and pre-built connectors save real development time; **RPA** (robotic process automation, simulating clicks and keystrokes) is the last resort for systems with no API and no low-code connector at all, brittle but sometimes the only option.

Tier	What to say
Passes	Reaches for whichever tool they personally know best, regardless of fit

Tier	What to say
Strong	The four-way decision framework above, correctly matching each option to what it's actually suited for
Extra points	+ [Business] RPA should be treated as a genuine last resort, not a default: it's the most fragile option, since it depends on a UI's exact layout never changing, and breaks immediately if a vendor redesigns their interface + [Edge cases] the same task can move between categories over time: an RPA solution built because no API existed should be revisited once a real API becomes available, rather than left in place purely out of inertia

Likely follow-ups: When would you choose a low-code tool over writing a script, even though you're capable of writing the script? What's the maintenance cost difference between these four options?

Red flag: defaulting to RPA or a macro for a task that a clean, available API would handle far more reliably. **Learn it in:** Chapter 19 (VBA/Apps Script) and Chapter 70, §70.5–70.6 (the same platforms, tested there for syntax, here for when to reach for them at all).

Rapid-fire, 78.1

#	Question	One-line answer	Extra point
Q78-002	What's the main maintenance risk of a low-code automation platform?	The logic lives inside a third-party tool's UI, often without version control, making it harder to review changes, roll back, or hand off to another person than a plain script would be	[Trade-offs] the speed of building in a low-code tool is a real, genuine advantage; the maintainability trade-off is the honest cost of that speed
Q78-003	Why might a company deliberately choose a script over a low-code tool even when both could do the job?	Version control, code review, and testing infrastructure the team already has for scripts, plus avoiding a recurring per-automation cost that many low-code platforms charge	[Business] the "cheaper to build" option (low-code) and the "cheaper to maintain at scale" option (a well-engineered script) are often different tools
Q78-004	What's the single biggest failure mode of an RPA-based automation?	Any change to the target application's UI, even a cosmetic one, silently breaks the automation, since it's simulating a specific sequence of clicks tied to a specific layout	[Real evidence] an RPA bot that clicks "the third button from the left" breaks the moment a fourth button gets added anywhere before it
Q78-005	How would you decide whether a task is even worth automating in the first place?	Weigh the time saved per run times how often it runs against the time it would take to build and maintain the automation, including the cost of it eventually breaking	[Business] a task done once a year, however tedious, is rarely worth automating; the same task done daily almost always is

78.2 Report and alert automation

Q78-006 · Design an automated daily MIS email sent to 200 managers, each seeing only their own team's numbers

Remember it as: One template, one query per recipient (or one query, filtered per recipient), not 200 separately hand-built reports.

Answer in one line: Build one parameterized report template and one data pull that includes every manager's team, then loop over the recipient list, filtering the data to each manager's own scope and rendering the same template with their specific numbers, rather than maintaining 200 separate report definitions.

Worked structure:

"One query pulls every manager's team performance in a single pass, tagged by manager ID. A loop iterates over the 200 managers, filters that single dataset down to each manager's rows, renders a shared HTML template with their specific numbers substituted in, and sends it via a transactional email service, not a personal inbox, so delivery is tracked and rate-limited sensibly. I'd log every send (recipient, timestamp, success or failure) so a manager who says 'I never got mine' can be checked against a real record instead of a guess, and I'd build in a retry for any individual send that fails, without blocking the other 199 emails from going out on time."

Tier	What to say
Passes	Describes building 200 separate reports or manually customizing each email
Strong	The one-template, one-filtered-pull, per-recipient-loop structure above
Extra points	+ [Edge cases] an individual send failure (a bad email address, a temporary mail server issue) shouldn't block the other 199, exactly the "partial failure isolation" principle from Chapter 77's pipeline design, applied here to email delivery instead of data loading + [Business] the send log is what turns "a manager complains they didn't get it" from a debugging mystery into a two-second lookup

Likely follow-ups: How would you handle a manager who should see two teams' data, not one? What would you monitor to know the whole batch actually completed successfully? **Red flag:** a design that fails the entire batch if any single recipient's send fails. **Learn it in:** Chapter 70, §70.5's

ExportAsFixedFormat /Outlook automation pattern, applied here at 200x scale with the failure-isolation discipline added.

Rapid-fire, 78.2

#	Question	One-line answer	Extra point
Q78-007	Why does an automated report need a "no news" state, not just "here's today's numbers"?	If a report silently sends nothing when there's genuinely nothing to report (versus when the pipeline feeding it failed), the recipient can't tell the difference between "all quiet" and "broken"	[Learn it in] Chapter 77, Q77-031 (the same "silence isn't automatically good news" principle)
Q78-008	What's the risk of sending an automated alert on every single anomaly, however small?	Alert fatigue (Chapter 77, Q77-032): recipients start ignoring the automation entirely, including the one alert that actually mattered	[Business] tune thresholds deliberately conservative for anything that emails a human directly, rather than flagging every minor fluctuation
Q78-009	Should a report automation retry immediately on failure, or wait?	Usually wait, with backoff (Q78-013): an immediate retry against a system that's down or overloaded often just adds to the load causing the failure in the first place	[Learn it in] §78.4 below

#	Question	One-line answer	Extra point
Q78-010	How would you let a stakeholder unsubscribe from an automated report without filing an IT ticket?	Build a simple self-service preference mechanism (a link, a form, a shared settings sheet) rather than requiring a developer to manually edit a recipient list every time	[Business] the absence of self-service unsubscribe is a common, avoidable source of stakeholders quietly filtering the automation to spam instead of asking for changes

78.3 APIs, webhooks, and reverse ETL

Q78-011 · What's the difference between polling an API and receiving a webhook, and when does each make sense?

Remember it as: *Polling is you repeatedly asking "anything new?" A webhook is the other system calling you the moment something happens, so you never have to ask.*

Answer in one line: **Polling** means repeatedly calling an API on a schedule to check for new data, simple to build but wasteful (most calls find nothing new) and inherently delayed by the polling interval; a **webhook** means the source system calls your endpoint the moment an event happens, near-instant and efficient, but requires the source to support webhooks at all and requires you to run a reliably available endpoint to receive them.

Tier	What to say
Passes	"Webhooks are more real-time" (true, no discussion of the trade-off or when polling is still the right choice)
Strong	The efficiency-vs-availability trade-off above, with a concrete case for each: polling suits a source with no webhook support, or where a few minutes of delay genuinely doesn't matter; webhooks suit anything needing near-real-time reaction
Extra points	+ [Edge cases] a webhook can be delivered more than once for the same event (network retries on the sender's side), so the receiving endpoint needs idempotent processing (Q78-012), exactly the same discipline Chapter 77 already teaches for pipeline loads + [Business] a webhook receiver going down, even briefly, can silently lose events forever unless the sender has its own retry/redelivery mechanism, worth confirming before relying on webhooks alone for anything critical

Likely follow-ups: How would you design a fallback in case your webhook receiver goes down for an hour? What's a webhook signature, and why does an endpoint need to verify one? **Red flag:** treating webhook delivery as guaranteed and exactly-once, with no idempotency handling. **Learn it in:** Chapter 77, Q77-008 (delivery guarantees, the identical concept applied to pipeline messages instead of webhooks).

Q78-012 · Build a webhook handler that safely processes a "lead created" event, even if the same event gets delivered twice

Remember it as: *Every webhook payload should carry its own unique event ID. Check it before processing, not after.*

Answer in one line: Check the incoming event's unique ID against a record of already-processed events before doing any work; if it's already been seen, acknowledge success and skip processing entirely, rather than reprocessing (and potentially duplicating downstream effects like creating a second CRM record).

Verified, live:

```
processed_ids = set()
def process_webhook(event_id, payload):
    if event_id in processed_ids:
        return "skipped (already processed)"
    processed_ids.add(event_id)
    return f"processed: {payload}"

process_webhook("evt_123", "lead created") # -> "processed: lead created"
process_webhook("evt_123", "lead created") # -> "skipped (already processed)" (same event,
delivered twice)
```

Tier	What to say
Passes	Processes every incoming webhook payload without checking for a duplicate delivery
Strong	The event-ID-check pattern above, correctly skipping a genuine duplicate
Extra points	+ [Validate] the real output above: the second identical delivery is correctly skipped, not silently reprocessed + [Business] in production, <code>processed_ids</code> would be a persistent store (a database table), not an in-memory set, since a set held only in memory is lost the moment the process restarts, silently reopening the exact duplicate-processing risk this pattern exists to close

Likely follow-ups: What HTTP status code should the webhook handler return, and why does that matter to the sender's retry behavior? How would you handle a webhook payload that fails processing partway through? **Red flag:** an in-memory-only duplicate check presented as a complete, production-ready solution with no mention of its persistence gap. **Learn it in:** Chapter 77, Q77-007 (idempotent upserts, the identical principle applied to a database load instead of an event handler).

Rapid-fire, 78.3

#	Question	One-line answer	Extra point
Q78-013	What's reverse ETL, and how is it different from normal ETL/ELT?	Normal ETL/ELT moves data <i>into</i> a warehouse for analysis; reverse ETL moves already-transformed, warehouse-computed data (like a churn score) <i>out</i> to operational tools (a CRM, an email platform) so business teams can act on it directly in the tools they already use	[Business] reverse ETL is what turns a model's output (Chapter 74) into something a sales rep actually sees inside their CRM, not just a table in a warehouse only analysts can query
Q78-014	Why does a reverse ETL sync into a CRM need to be an upsert, not a plain insert?	Re-running the sync (daily, say) would otherwise create a duplicate CRM record for every lead on every single run, instead of updating the existing one's score	[Validate] verified live: syncing <code>lead_101</code> 's updated score correctly updated the existing record (72 → 80) rather than creating a second one; the total record count stayed accurate at 3 after both a new and an existing lead were synced

#	Question	One-line answer	Extra point
Q78-015	What's an API rate limit, and why does an integration need to respect it proactively?	A cap on how many requests a system will accept in a given time window; exceeding it usually gets your requests rejected or your access temporarily blocked, which is worse for the automation than simply pacing itself	[Validate] verified live: a simple rate limiter correctly allowed 3 rapid calls against a 3-per-second cap and rejected the 4th and 5th within that same window
Q78-016	What's an API key vs. OAuth, at a level useful for an integration decision?	An API key is a single, simple secret string identifying the calling application; OAuth is a more complex flow granting scoped, revocable, user-specific access without ever sharing the user's actual password	[Business] OAuth is the right choice whenever an integration needs to act <i>on behalf of a specific user</i> , not just as an application in general

78.4 Failure handling and monitoring for automations

Q78-017 · An automation calls a flaky third-party API that sometimes fails transiently. Build retry logic that doesn't hammer the failing service

Remember it as: *Retrying immediately just adds load to a service that's already struggling. Waiting longer after each failure gives it room to recover.*

Answer in one line: Use **exponential backoff**: retry after a short delay, and double that delay after each subsequent failure, up to a maximum number of attempts, so a transient blip gets a quick retry while a sustained outage doesn't get hammered with rapid-fire requests that make the underlying problem worse.

Verified, live:

```
def call_with_retry(func, max_attempts=5, base_delay=0.01):
    for attempt in range(1, max_attempts + 1):
        try:
            return func()
        except ConnectionError:
            if attempt == max_attempts:
                raise
            delay = base_delay * (2 ** (attempt - 1))
            time.sleep(delay)
```

```
attempt 1 failed, retrying in 0.010s
attempt 2 failed, retrying in 0.020s
succeeded on attempt 3
```

Tier	What to say
Passes	Retries immediately in a tight loop with no delay at all
Strong	The exponential backoff pattern above, with a maximum attempt cap so it eventually gives up rather than retrying forever

Tier	What to say
Extra points	+ [Validate] the real run above: two failures, each with a longer wait than the last, then success on the third attempt + [Edge cases] a real implementation should add jitter (a small random amount added to each delay) so that if many clients are retrying the same recovering service at once, they don't all retry at exactly the same synchronized moment and cause a new spike

Likely follow-ups: What's the difference between a transient error worth retrying and a permanent one that isn't? How would you decide the maximum number of attempts? **Red flag:** a retry loop with no delay at all, or no maximum attempt limit, risking an infinite retry loop. **Learn it in:** Chapter 77, §77.2 (delivery guarantees and idempotency, the surrounding concepts this retry logic depends on to be safe).

Q78-018 · An automation that's run reliably for months silently stopped working last week. Walk through how you'd find out, and how you'd prevent it happening silently again

Remember it as: "It stopped running" and "it ran but did nothing useful" are two different failures, and the second one is far more dangerous because nothing looks broken from the outside.

Answer in one line: First check whether it's actually still executing at all (a scheduler log, a "last run" timestamp) versus running but silently failing partway through or producing empty/wrong output; then check what changed around the time it stopped (an expired credential, an upstream API change, a permissions change); then, regardless of the specific cause found, add a monitoring check so the next silent failure surfaces immediately instead of a week later.

Worked structure:

"I'd check the scheduler first: did it even attempt to run? If yes, I'd check its logs for the actual error, most commonly an expired API token or OAuth credential, a changed API response format, or a permissions change on the target system. If it ran with no error but produced no visible effect, I'd suspect a silent logic issue, like a filter condition that now matches zero rows because of an upstream data change. Either way, once I've found and fixed the specific cause, I'd add a monitoring check (Chapter 77's volume-anomaly pattern, applied here to an automation instead of a data pipeline): alert if this automation hasn't successfully completed within its expected window, rather than relying on someone noticing its absence."

Tier	What to say
Passes	Investigates and fixes the immediate cause with no mention of preventing a future silent recurrence
Strong	The two-stage diagnosis above (did it run at all vs. did it run but do nothing useful), plus a monitoring fix to close the loop
Extra points	+ [Business] explicitly distinguishes "stopped executing" from "executed but silently did nothing," since the second is a genuinely harder, more dangerous failure mode to notice, exactly Chapter 77's "quiet day" story, now applied to a business-facing automation instead of a data pipeline

Likely follow-ups: What's the most common real-world cause of an automation silently failing after months of working fine? How would you design the monitoring alert to avoid false positives on a legitimately quiet day (a holiday, say)? **Red flag:** fixing the immediate cause with no monitoring added

to catch the next occurrence automatically. **Learn it in:** Chapter 77, Q77-028 and Q77-031 (the identical monitoring discipline, here applied to automations rather than data pipelines).

Rapid-fire, 78.4

#	Question	One-line answer	Extra point
Q78-019	What's the most common real cause of a previously-working automation breaking without any code change?	An expired credential, API key, or OAuth token; these have finite lifespans and their expiry is rarely tied to any code deployment, so nothing in version control explains the sudden failure	[Business] tracking credential expiry dates proactively (a calendar reminder, a monitoring check) avoids being surprised by this specific, very common failure mode
Q78-020	Why should an automation log its own successful runs, not just its failures?	Without a positive "last successful run" record, there's no way to distinguish "silently broken for a week" from "just ran successfully an hour ago"	[Learn it in] Chapter 77, Q77-031
Q78-021	What's a circuit breaker pattern, in automation terms?	After a certain number of consecutive failures calling an external system, stop trying entirely for a cooldown period, rather than continuing to retry a system that's clearly down	[Trade-offs] trades a short window of "giving up early" against not making a struggling system's outage worse by continuing to hammer it
Q78-022	Should an automation's failure alert go to an individual person or a team channel?	Generally a team channel or shared on-call rotation, not one individual, so the automation doesn't silently go unmonitored the day that one person is on leave	[Business] a single point of alerting failure is exactly as fragile as a single point of automation failure

78.5 Basic-but-tricky automation questions

Q78-023 · Does automating a broken manual process actually fix it?

Remember it as: Automating a bad process just makes it fail faster and more consistently. Fix the process, then automate it, not the other way around.

Answer in one line: No: automating a process that's fundamentally flawed (wrong logic, unclear ownership, missing edge-case handling) usually just executes the same flaw faster and more consistently, and can make the underlying problem *harder* to notice, since a human doing the task manually might catch an obviously wrong result before acting on it, while an automation won't.

Tier	What to say
Passes	Assumes automating a manual process is always a pure improvement
Strong	Explicitly names the risk: automation removes the human sanity-check step that was, often invisibly, catching errors before
Extra points	+ [Business] the right sequence is almost always: understand and fix the process first, <i>then</i> automate the corrected version, not automate first and hope problems surface on their own

Likely follow-ups: How would you know, before automating, whether a manual process has a hidden human sanity-check step worth preserving? What would you add to the automated version to replace that lost check? **Red flag:** treating "we automated it" as inherently synonymous with "we improved it." **Learn it in:** Chapter 76B, §76.4's as-is-before-to-be discipline, the same principle applied to automation instead of process redesign.

Rapid-fire, 78.5

#	Question	One-line answer	Extra point
Q78-024	Is "no-code" the same thing as "no technical skill required"?	No: a no-code tool removes the need to write syntax, but building a genuinely reliable automation in one still requires the same logical thinking (edge cases, error handling) a script would need	[Business] a poorly-designed no-code automation fails in exactly the same ways a poorly-designed script does, just without a stack trace to help debug it
Q78-025	Why might a "quick win" automation built by a single enthusiastic employee become a long-term liability?	If it's undocumented and only that one person understands it, the automation becomes a single point of failure the moment they leave or go on leave, exactly the same risk as an undocumented process	[Real evidence] this is one of the most common real sources of "nobody knows why this stopped working" incidents
Q78-026	Should every manual task with any repetition at all be automated eventually?	No: some tasks are rare enough, or variable enough in their exact steps, that the automation's build and maintenance cost never pays back the time saved	[Learn it in] Q78-005
Q78-027	What's the risk of over-customizing an automation for one specific stakeholder's exact preferences?	It becomes fragile and hard to extend to a second stakeholder with slightly different needs, and often duplicates logic that should have been one shared, parameterized design	[Learn it in] Q78-006's one-template, per-recipient-filter pattern is the direct antidote to this

78.6 Full design cases

Q78-028 · Design case: automate a daily MIS email for 200 managers, each seeing only their own team's numbers

(Fully worked in §78.2, Q78-006 above; referenced here as this chapter's first complete design case per the blueprint's own four named cases.)

Q78-029 · Design case: sync lead scores into the CRM without creating duplicates

What they're really testing: whether the upsert discipline from Chapter 77 gets applied correctly to a live, operational sync target, not just a data warehouse table.

Talked through live: "Each lead needs a stable external ID shared between the scoring system and the CRM, either the CRM's own record ID if the scoring system already knows it, or an agreed common key like email. The sync checks for an existing record by that ID: if found, update its score field; if not,

create a new record. This is exactly the upsert pattern from Chapter 77's idempotency work, just targeting a CRM's API instead of a database table."

Verified, live:

```
existing_crm_records = {"lead_101": {"score": 72}, "lead_102": {"score": 55}}
sync_lead_score("lead_101", 80) # -> "updated lead_101: 72 -> 80"
sync_lead_score("lead_103", 60) # -> "created lead_103 with score 60"
# record count after both operations: 3, correctly, no duplicate created for lead_101
```

Extra-points moves demonstrated: [Depth] explicitly connected this to Chapter 77's upsert concept rather than treating it as a new problem. [Edge cases] named the stable-external-ID requirement as the actual precondition for this working at all, not an afterthought. [Validate] the real record count (3, not 4) proves no duplicate was created.

Likely follow-ups: What would you do if the CRM's API doesn't support a native upsert operation, only separate create and update calls? How would you handle a lead that exists in the CRM but was deleted from the source scoring system? **Red flag:** a sync design using a plain "create" call on every run, with no check for an existing record first. **Learn it in:** Chapter 77, Q77-007 (idempotent upserts, the source of this exact pattern).

Q78-030 · Design case: an automation silently stopped working last week. How do you find out, and how do you prevent it happening silently again?

(Fully worked in §78.4, Q78-018 above; this chapter's third named design case.)

Q78-031 · Design case: choosing between macros, scripts, Python, and low-code tools for a specific new request

(Fully worked in §78.1, Q78-001 above, applied generally; the specific decision framework there is designed to be reused for any concrete version of this request a real interview might pose.)

Common mistakes and how to spot them

Mistake	Symptom	Fix
Defaulting to RPA or a macro regardless of fit	A fragile automation where a clean API would have worked	Match the tool to how the task actually touches its systems (§78.1)
An immediate, no-delay retry loop against a flaky API	Adds load to an already-struggling service, can worsen an outage	Exponential backoff with a maximum attempt cap
A webhook handler with no duplicate-event check	A duplicate delivery creates a duplicate downstream record	Check a unique event ID before processing, persisted, not in-memory only
A CRM sync using plain "create" every run	Duplicate records pile up on every sync cycle	Upsert by a stable external ID
No monitoring for an automation's own silent failure	A break goes unnoticed for days or weeks	Alert on absence of a successful run, not just on explicit errors

Mistake	Symptom	Fix
Automating a broken process as-is	The flaw executes faster and more consistently, harder to notice	Fix the process first, then automate the corrected version
One person quietly owning an undocumented "quick win" automation	A single point of failure when that person is unavailable	Document ownership and logic; don't let critical automation live in one person's head

In the real world: the automation that broke because of a renamed column

Meera is asked in an interview to describe debugging a real automation failure. She describes a weekly commission-calculation automation that had run correctly for over a year, then started producing wrong numbers, not failing outright, just quietly wrong, which took two pay cycles to notice because nobody was checking the automation's output against a manual calculation anymore, exactly the trust that made the automation valuable in the first place also made its silent failure harder to catch.

She traces it to a source spreadsheet where someone had renamed a column from "Discount %" to "Discount Percentage," and her script had been reading that column by its exact name. The rename didn't break the script outright, since the script's error handling silently defaulted a missing column to zero rather than crashing, meaning discounts were being read as zero for every single row for two full pay cycles, quietly inflating every calculated commission.

The interviewer's follow-up: what did she change? Her answer: two things, not one. First, the immediate fix, reading the column by position with a name-validation check that now fails loudly if the expected column name isn't found, rather than silently defaulting. Second, and the part she says mattered more: "I stopped treating 'the automation ran without an error' as proof it worked correctly. Now it validates its own output against a rough expected range before sending anything downstream, so a silently wrong result gets caught the same day, not two pay cycles later." That's this chapter's whole method: the fix for the specific bug matters less than the systemic change that catches the next silent failure automatically.

Tools

Python for scripted automation and API integration; **VBA/Apps Script** (Chapter 19, Chapter 70) for in-application macros; common low-code platforms (Zapier, Power Automate, Make) for cross-system automation without custom code; a scheduler (cron, Airflow, or a platform's built-in scheduling) for anything running on a timer. Every code pattern in this chapter runs on plain Python 3.12 with no special libraries beyond the standard library.

The project

Goal: apply this chapter's method to a real automation of your own.

1. Pick one manual, repetitive task from your own work and run it through Q78-001's decision framework: which of the four approaches actually fits, and why.
 2. Add retry-with-backoff logic to one real script you have that calls an external system, and test that it actually waits longer between successive failures.
 3. Add a duplicate-event or duplicate-record check to one integration you maintain, and prove (the way Q78-012 and Q78-029 did) that running it twice doesn't create a duplicate.
 4. Write the monitoring check you'd add to catch your most critical automation silently failing, and say specifically what "silently failing" would look like for that one.
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Final-week revision list

Q78-001, Q78-006, Q78-011, Q78-012, Q78-013, Q78-017, Q78-018, Q78-023, Q78-029.

Key terms

macro vs. script vs. low-code vs. RPA · report automation · alert fatigue · polling vs. webhook · webhook idempotency · reverse ETL · API rate limit · API key vs. OAuth · exponential backoff · jitter · circuit breaker · silent failure · monitoring (absence of success) · upsert (in an integration context) · single point of failure (undocumented automation)

Where this leads

- **Chapter 69, The Extra-Points Method**, is the rubric and move set every answer above is written against.
- **Chapter 77, Data Engineering & Data System Design Bank**, already covers idempotency, retries, and monitoring from the data-pipeline angle; this chapter applies the identical discipline to business-facing automation and integrations.
- **Chapter 70, §70.5–70.6**, covers the VBA/Apps Script syntax this chapter's tool-choice framework (§78.1) helps decide when to actually reach for.
- **Chapter 76B, §76.4**, supplies the as-is-before-to-be discipline this chapter's Q78-023 reuses directly for automation specifically.